IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

STABILITY AND CONTROL OF NON-LINEAR SYSTEMS

Wednesday, 13 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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1. Consider the following nonlinear dynamical system:

$$\begin{aligned} \dot{x}_1 &= -x_2 + |x_1 + 1| - |x_1 - 1| - x_1 \\ \dot{x}_2 &= -x_2^3 + 2x_2 + x_1 \end{aligned}$$

Let $x = [x_1, x_2]'$ denote the state of the system, evolving in $\mathbb{R}^2$, and $f(\cdot) : \mathbb{R}^2 \to \mathbb{R}^2$ denote the corresponding vector-field.

a) Analyze the continuity properties of $f(\cdot)$ and discuss their implications on existence and uniqueness of solutions of the associated system of differential equations. [2]

b) Describe analytically and sketch graphically the nullclines of the vector-field (try to be accurate for best results). [2]

c) Find regions in $\mathbb{R}^2$ delimited by nullclines and identify the orientation of the vector-field in each one of them. [3]

d) Find, if any, forward invariant regions among those identified in part c) (Justify your answer). [2]

e) Compute graphically all equilibria of the system. [2]

f) Linearize the system around each equilibrium and discuss its local phase portrait. [3]

g) Show that the box $B = \{x : |x_1| \leq 4, |x_2| \leq 2\}$ is forward invariant. [3]

h) Take $x_0 = [1, 1]'$. How do you expect $\omega(x_0)$ to look like ? (*Hint*: Look for a forward invariant set which does not contain equilibria). [3]

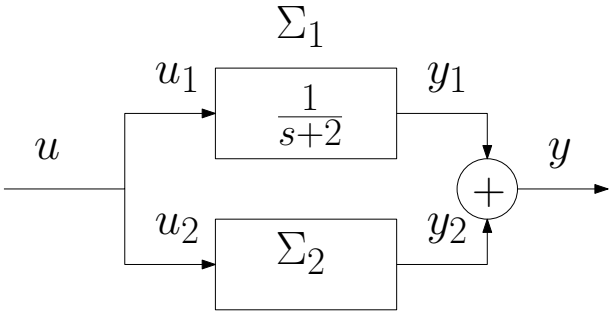


Figure 2.1 Parallel systems' interconnection

2. Consider the parallel interconnection of systems Σ_1 and Σ_2 shown in Fig. 2.1. The first subsystem, Σ_1, of input u_1 and output y_1 is linear, one-dimensional and described by the Input/Output transfer function $1/(s+2)$. The second subsystem, Σ_2, is two-dimensional, described by the following nonlinear second order differential equation:

$$\ddot{z} = -\dot{z}^5 - \sin(z) + u_2$$

and output equation: $y_2 = \dot{z}$.

a) Choose state variables for each subsystem Σ_i, $i = 1,2$ and write down, for each one, a state-space representations of the form $\dot{x}_i = f_i(x_i, u_i), \quad y_i = h_i(x_i)$, for suitable functions f_i and h_i. [4]

b) Show that system Σ_1 is passive from u_1 to y_1. [2]

c) Find a suitable storage function $S_2(x_2)$ and show that subsystem Σ_2 is passive from u_2 to y_2 [4]

d) Consider next a periodic input $u(\cdot)$ of period $T > 0$, and assume that for some suitable initial condition the corresponding output $y(\cdot)$ and state $x(\cdot)$ solutions of the parallel interconnection are also periodic of period T. Show that necessarily:

$$\int_0^T u(t)y(t)\,dt \geq 0.$$

[4]

e) Consider next the subsystem Σ_2 in isolation. By using Lyapunov analysis show that for any positive gain $k > 1$, the negative feedback interconnection $u_2 = -kz$ results in a globally asymptotically stable system. [6]

3. Consider the following class of functions

$$\mathscr{G} := \{g : \mathbb{R} \to \mathbb{R} : g(0) = 0, g(\cdot) \text{ is increasing, odd and Lipschitz}\}.$$

Let g_1 and g_2 be arbitrary functions in $\mathscr{G}$. We define the following two-dimensional system of differential equations:

$$\begin{array}{rcl} \dot{x}_1 & = & -g_1(2x_1) + g_1(x_2) \\ \dot{x}_2 & = & -g_2(2x_2) + g_2(x_1) \end{array} .$$

a) Recast the previous equations as a feedback interconnection of two scalar subsystems. Identify their input signals, output signals and state-space representations. [4]

b) Show that the x_1 and x_2 subsystem previously identified are both Input-to-State stable with respect to their input signals. Compute the associated Input-to-State gain, trying to be as tight as possible in your estimates. [6]

c) Argue that their feedback interconnection, i.e. the considered closed-loop system, is globally asymptotically stable. [2]

d) Consider now the n scalar systems of equations:

$$\dot{x}_i = -g_i(2x_i) + g_i(u_i), \qquad y_i = x_i$$

for $i = 1, \ldots, n$ and $g_i \in \mathscr{G}$. Consider their series interconnection (in order of increasing index i), and denote its input $u = u_1$ and output $y = y_n$. We define as the Input-Output Asymptotic Gain, any $\mathscr{K}_\infty$ function γ such that:

$$\limsup_{t \to +\infty} |y(t)| \leq \gamma(\limsup_{t \to +\infty} |u(t)|).$$

for all input signals $u(\cdot)$ and all corresponding output signals $y(\cdot)$. Can you use the previously derived Input-to-State gains to compute a suitable expression for $\gamma(\cdot)$? (Justify your answer). [4]

e) Consider next the feedback $u = Ky$. Find, for each value of n (number of subsystems in the loop), an estimate for the range of K that makes the interconnection Globally Asymptotically Stable at the origin. (Hint: a generalized small gain theorem applies using Input-Output gains alone and for an arbitrary number of subsystems). [4]

4. Consider the nonlinear control system of equations:

$$\begin{aligned}
\dot{x}_1 &= \arctan(x_1) + u(\alpha + \cos(x_2)) \\
\dot{x}_2 &= x_1 - x_2 + x_3 + \arctan(x_1) + u(\alpha + \cos(x_2)) \\
\dot{x}_3 &= x_1 - x_2 - x_3 + x_1^3 \\
\\
y &= x_1
\end{aligned}$$

with state $x = [x_1, x_2, x_3]' \in \mathbb{R}^3$, and scalar input and output variables u and y, respectively. Finally, $\alpha \in \mathbb{R}$ is a constant parameter.

a) Discuss the relative degree of the system at the origin (as a function of α) and justify whether this is locally defined, globally defined or undefined. [4]

b) Find a state-feedback $u = k(\alpha, x, v)$ so that the resulting closed-loop system is linear from the auxiliary input $v \in \mathbb{R}$ to output y. [3]

c) Design a stabilizing linear feedback control action $v = k_1 y$ so that the Input-Output dynamics and the equilibrium corresponding to $y(t) \equiv 0$ are globally asymptotically stabilized in closed-loop. [2]

d) Bring the system in normal form and highlight the internal dynamics. [4]

e) Assess whether the internal dynamics are Input-to-State Stable (ISS) with respect to their associated input signal y. [3]

f) Discuss the stability of the overall closed-loop system in the light of the ISS analysis carried out in the previous question (Hint: how does α affect your conclusions ?). [4]